

Governing Patient-Facing AI-Generated Video in Digital Health

1. Key Terminology & Scope

- **Patient-Facing AI-Generated Video:** Synthetic or AI-mediated video presenting realistic human likeness, voices, or clinical communication cues that influence patient trust or comprehension.
- **Operational Governance:** Local institutional processes through which these systems are reviewed, approved, monitored, and re-evaluated over time.
- **The Governance Gap:** High-level national laws and regulations (e.g., the EU AI Act) exist, but they lack the granular instructions local institutions need to translate broad concepts (transparency, consent, equity) into case-level deployment decisions. The proposed matrix acts as an **operational layer** to bridge this gap.

2. Core Framework: The Risk-and-Ethics Matrix

The framework cross-tabulates **Residual Clinical Risk** against an **Ethical Alignment Score (EAS)** to yield one of four actionable deployment dispositions:

Axis A: Residual Clinical Risk (1 to 4 Tiers)

Calculated as **Likelihood times Severity** *after* specific institutional safeguards (e.g., clinician script reviews, constrained generation) are applied:

- **Likelihood:** Rare, Unlikely, Possible, Likely.
- **Severity:** Negligible, Minor (transient confusion), Major (consequential misinformation/psychological distress), Catastrophic (severe injury/system-level misinformation).
- **Resulting Tiers:** Low, Moderate, High, or Extreme.

Axis B: Ethical Alignment Score (EAS: 0 to 8 Points)

Operationalizes the four core bioethical principles. Each principle is scored from 0 (violated) to 2 (clearly upheld):

1. **Autonomy:** Transparency of AI use, accuracy of identity representation, and presence of an opt-out.
2. **Beneficence:** Credible, evidence-informed patient benefit proportionate to burdens.
3. **Nonmaleficence:** Robust mitigations (human-in-the-loop, crisis plans) minimizing psychological or data harms.

4. **Justice:** Equitable access, bias mitigation, and accessible complaint pathways.
- **EAS Banding:** High (7–8), Medium (4–6), Low (0–3).

3. The Four Deployment Dispositions & Minimum Controls

Disposition	Entry Rules (Risk + EAS)	Required Minimum Controls
Encourage	Low Risk + High EAS	Routine quality assurance and clear content disclosure.
Permit with Oversight	Moderate Risk + $\geq$ Medium EAS <i>or</i> Low Risk + Medium EAS	Human-in-the-loop review, strict audit trails, time-limited approvals, and post-deployment monitoring.
Restrict or Redesign	High/Extreme Risk <i>or</i> Low EAS (without intrinsic deception)	Scope narrowing, mandatory safety guardrail upgrades, and protocolized pilot testing before rescoreing.
Prohibit	Extreme Risk <i>or</i> Low EAS with intrinsic deception/manipulation	Disallowed entirely; immediate takedown, platform reporting, and legal remedies.

4. Operational Lifecycle Workflow

The institutional workflow consists of a repeatable, four-step iterative loop:

[Use Case Dossier Submitted] → [Independent Triad Panel Scoring] → [Governance Disposition Decision] → [Post-Deployment Monitoring & Rereview]

1. **Dossier Submission:** Proponents outline the generation pipeline, target audience, and risk mitigation log.
2. **Triad Panel Evaluation:** An interdisciplinary panel consisting of a **clinician**, a **bioethicist**, and an **IT/Safety lead independently** scores and reconciles the matrix.
3. **Post-Deployment Monitoring Set:** Institutions must continuously track:
 - *Comprehension:* Users teach back checks and report confusion.
 - *System Burden:* Content-attributable follow-up calls or messages.
 - *Equity Gaps:* Stratification of complaints/comprehension by age, language, and literacy.
4. **Rereview Triggers:** A procedural reassessment is automatically triggered by material updates (model, prompt template, or language guidelines), signal excursions (spikes in user confusion), or equity flags.

5. Illustrative Case Stabilities mapped in the Paper

- **Case 1: Multilingual Patient Education Avatar via Portal:** Clinician-vetted scripts, transparent labeling. *Result:* **Encourage** (Low risk, High EAS).
- **Case 2: "Deepfake Therapy" for Grief Support:** Specialist-led simulation using a deceased relative's voice/likeness under strict protocols. *Result:* **Permit with Oversight** (Moderate-high risk, Medium EAS).
- **Case 3: Impersonated Physician Endorsements on Social Media:** Cloned clinician likeness promoting unverified health products without consent. *Result:* **Prohibit** (Extreme risk, Low EAS due to intrinsic deception).

6. Implementation Architecture: Global vs. Local Tuning

To remain scalable across various regulatory landscapes, the framework divides its components into strict standards and flexible settings:

- **Core Standardized Elements (Non-negotiable):** The 4-tier risk logic, the 0-2 EAS principle scoring method, the 4 disposition categories, and the mandatory requirement for human oversight and versioned registries.
- **Locally Tunable Parameters (Adaptable):** The specific layout of the dossier, language levels used in disclosure wording, specific statistical indicator thresholds, and the option for resource-limited settings to use a simplified two-reviewer pathway instead of the full triad panel.